

Project 5: Final Capstone – AI-Enabled SaaS Product

Duration: 25 June 2026 – 30 June 2026

Deadline: 30 June 2026

1. Project Overview

This is the **final project of the internship program**. In this phase, you are expected to combine everything you have learned:

- Python core engineering
- Backend development using Django
- Frontend development using Next.js
- AI/NLP concepts

You will build a **production-level SaaS (Software as a Service) application** that integrates backend, frontend, and AI into one complete system.

This project simulates a real-world product development scenario where you are responsible for delivering a usable and deployable application.

2. Problem Statement

Modern applications are expected to:

- Provide a seamless user experience
- Handle backend logic efficiently
- Integrate intelligent features using AI
- Be scalable and production-ready

Your task is to design and build a complete system that solves a real-world problem using a full-stack and AI-driven approach.

3. Project Objectives

By completing this project, you should be able to:

- Design and build an end-to-end software system
- Integrate backend, frontend, and AI components
- Apply best practices in system design and architecture
- Deploy and present a real product
- Demonstrate ownership of a complete project lifecycle

4. Project Options (Choose One)

You must select **one of the following options** and build it completely.

Option A: AI SaaS Dashboard

Description

Build a business dashboard that provides insights and predictions based on data.

Example Features

- Upload business data (CSV/Excel)
- Visualize key metrics (charts, graphs)
- AI-based predictions (e.g., sales forecasting)
- Summary insights

Option B: Smart Recruitment System

Description

Extend your previous resume analyzer into a full system for recruiters.

Example Features

- Resume upload and analysis
- Candidate scoring and ranking
- Dashboard for recruiters
- Job description matching

Option C: E-commerce Platform with AI Recommendations

Description

Build an online store with intelligent product recommendations.

Example Features

- Product catalog
- User authentication
- Cart and order system
- AI-based recommendation engine
 - “Recommended for you”
 - “Similar products”

5. Mandatory Features

Regardless of the option you choose, your system must include the following:

5.1 Backend System

- Build REST APIs using Django
- Proper database design
- Secure endpoints

5.2 Frontend Application

- Build UI using Next.js
- Clean and responsive design
- Proper API integration

5.3 AI Integration

- Include at least one AI feature:
 - Prediction
 - Recommendation
 - Text analysis
- Clearly explain how your AI works

5.4 Authentication System

- User signup/login
- Secure access to features
- Token-based authentication\

5.5 Deployment (optional)

- Your application must be live
- Use platforms such as:
 - Vercel (frontend)
 - Render / Railway / AWS (backend)

6. Advanced Requirements

6.1 Clean Architecture

- Proper separation of concerns:
 - Models
 - Views
 - Services
 - Utilities

6.2 Scalable Structure

- Code should be easy to extend
- Avoid hardcoding
- Follow best practices

6.3 Proper Error Handling

- Handle API errors gracefully
- Show meaningful messages to users
- Log errors properly

7. Deliverables

By the deadline, you must submit:

7.1 Live Application (optional)

- Fully deployed and accessible online
- All core features working

7.2 Complete Documentation

Your documentation must include:

- Project overview
- Architecture explanation
- Tech stack used
- Setup instructions
- API documentation
- Explanation of AI logic

7.3 Final Presentation

You must present your project covering:

- Problem you solved
- System architecture
- Demo of features
- AI implementation
- Challenges faced

7.4 GitHub Repository

- Clean and structured code
- Proper commit history
- README file

8. Evaluation Criteria

Your project will be evaluated based on:

8.1 End-to-End Integration

- Does the system work as a complete product?
- Are frontend, backend, and AI properly connected?

8.2 System Design

- Is the architecture well thought out?
- Is the database properly designed?

8.3 Production Readiness

- Is the application stable?
- Can real users use it?
- Is it properly deployed?

8.4 Code Quality

- Clean, readable, and maintainable code
- Proper structure and organization

8.5 Innovation and Effort

- Creativity in features
- Depth of implementation
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09. Final Outcome Expectations

By the end of this program, you are expected to be able to:

- Build production-level backend systems
- Develop complete full-stack applications
- Apply AI in real-world scenarios
- Deploy and demonstrate working products

10. Important Instructions

- Treat this like a real product, not an assignment
- Focus on quality, not just completing features
- Test your system thoroughly
- Ensure your deployment is stable
- Be prepared to explain every part of your system